19. Implement the Erosion Morphological Operation Using OpenCV in Python

CODE:

```
import cv2
import matplotlib.pyplot as plt

# Read the input image
img = cv2.imread('image.jpg')

if img is None:
    print("Error: Image not found!")
else:
    # Convert the image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Convert grayscale image to binary image
    _, binary = cv2.threshold(
        gray, 127, 255, cv2.THRESH_BINARY
    )

    # Create a structuring element (kernel)
    kernel = cv2.getStructuringElement(
        cv2.MORPH_RECT, (5, 5)
    )

    # Apply erosion
    erosion = cv2.erode(
        binary, kernel, iterations=1
    )

    # Display the images
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(binary, cmap='gray')
    plt.title('Binary Image')
    plt.axis('off')

    plt.subplot(1, 3, 3)
    plt.imshow(erosion, cmap='gray')
    plt.title('Eroded Image')
    plt.axis('off')
```

```
plt.tight_layout()
plt.show()

# Save the eroded image
cv2.imwrite('eroded_image.jpg', erosion)

print("Erosion applied successfully!")
```

